

Examination System Brief

Tables

- **Roles:** Stores user access levels (Admin, Instructor, Student).
- **Branches / Departments / Intake / Track:** Hierarchical structures for organization.
- **Track_Intake / Branch_Track:** Junction tables for many-to-many relationships.
- **Users:** Core authentication table for all system participants.
- **Instructor / Student:** Profile tables linked to the Users table.
- **Course:** Stores academic subject details and passing thresholds.
- **Question / Choice:** Bank of questions and their possible MCQ answers.
- **Exam:** Defines exam instances, timings, and associated courses.
- **Question_Exam:** Junction table defining which questions appear in which exam.
- **Student_Ans:** Stores specific student responses and automated grading results.
- **Student_Course / Student_Exam:** Tracks enrollment and exam participation.
- **Instructor_Course:** Assigns instructors to specific course instances.

Functions

- **GetStudentExamDegree:** Calculates the sum of degrees for a student in a specific exam.
- **FN_CalcTotalCourseDegree:** Converts raw exam scores into a percentage based on MaxDegree.
- **FN_CheckTextAnswer:** Uses pattern matching to provide partial credit for text-based answers.
- **FN_GetStudentStatus:** Returns 'Passed' or 'Failed' based on comparing results to MinDegree.

Stored Procedures

- **SP_AddStudent / SP_AddInstructor / SP_AddCourse:** Standardized entry of core entities.
- **SP_AddQuestion:** Inserts questions into the bank for a specific course.

- **CreateExam:** Initializes a new exam record with timing and metadata.
- **GenerateExamQuestions:** Randomly pulls a set number of questions into an exam.
- **AssignExamToStudents:** Bulk inserts exam records for all eligible students in a track/branch.
- **SubmitAnswer:** Records a student's answer and performs an immediate correctness check.
- **CalculateExamResult:** Aggregates and returns the total score for a finished exam.
- **SP_StartExam:** Validation logic to ensure a student is eligible and within the time window.

Views

- **V_StudentResults:** Summary of student names and their total exam scores.
- **V_ExamDetails:** Maps exams to their specific questions and assigned points.
- **VW_StudentAnswer:** An encrypted view showing students' actual answers vs. correctness.
- **V_InstructorPerformance:** Reports on instructors and the volume of students they teach.
- **V_ExamAnalytics:** Provides high-level Pass/Fail statistics per exam.
- **VW_StudentScheduleExams:** Lists upcoming exams for students based on current time.

Triggers

- **TR_CheckDegree (Instead Of):** Ensures total exam points do not exceed the course's MaxDegree.
- **TR_CheckAnswer (After):** Automatically updates IsCorrect and Degree upon answer insertion.
- **TR_AutoCalculateResults (After):** Calculates final percentages and letter grades (A-F).